Prediction of Two-Phase Flow Rate Through Wellhead Chokes in Oil Wells: Overcoming Domain Collapse and Physical Inconsistencies in Virtual Flow Metering

Academic Project Report

Machine Learning & Multiphase Reservoir Analytics

Stanford CS229 Benchmark Replication & Physics-Constrained Extension

Abstract

The wellhead choke valve is a critical element of surface petroleum production architecture. It controls multiphase reservoir flux, suppresses transient pressure surges, and protects downstream separation facilities. Reliable estimation of multiphase flow rates through chokes is essential for continuous reservoir surveillance, production allocation, and real-time field optimization. Because physical multiphase flow meters (MPFMs) incur high capital expenditure ($\$100{,}000 - \$300{,}000$ per wellhead) and frequently suffer operational failure in abrasive field conditions, software-driven Virtual Flow Metering (VFM) has gained strong industry traction. Prior benchmark investigations, notably by Nazari and Alshafloot at Stanford University, demonstrated that classical empirical choke correlations fail when applied across broad field datasets, and proposed ensemble decision trees (specifically Extra Trees regressors) as optimal alternatives ($R^2 = 0.58$, Pearson $r = 0.69$).

However, standard machine learning formulations suffer from three critical deficiencies:

1. High reported performance ($R^2 > 0.95$) is frequently an artifact of temporal data leakage induced by naive random train/test splits across autocorrelated daily production records;
2. Models exhibit severe domain collapse under cross-well or cross-field transfer ($R^2 < 0.50$); and
3. Unconstrained decision boundaries produce non-physical predictions, such as predicting decreasing flow rates as the choke orifice is throttled open.

In this work, we resolve these vulnerabilities using high-rate offshore production telemetry

from the Equinor Volve Field ($6{,}658$ steady-state production days across four wellbores). We construct a physics-engineered feature pipeline grounded in Euler-Bernoulli orifice dynamics and acoustic critical pressure thresholds, combined with a Monotonic Physics-Constrained Gradient Boosting architecture (LightGBM). Under rigorous blind cross-well transfer (training on well 15/9-F-14 and testing on unseen well 15/9-F-12), our proposed model improves the determination coefficient from $R^2 = 0.456$ to $R^2 = 0.748$ and reduces the Mean Absolute Percentage Error (MAPE) from 38.67% to 24.25%. Furthermore, in synthetic throttle sweeps, our model completely eliminates non-physical rate drops (0 monotonicity violations versus 23 violations in the Stanford Extra Trees baseline).

1. Introduction and Background

In oil and gas production systems, fluids produced at the surface wellhead consist of complex, turbulent multiphase mixtures of crude oil, formation water, and associated gas. Regulating this production requires wellhead chokes—calibrated mechanical orifices that impose controlled backpressure on the formation:

- **Reservoir Conservation:** Preventing premature gas or water coning and sand influx caused by excessive drawdown.
- **Surface Protection:** Mitigating pressure pulsations and decoupling high reservoir pressures from low-pressure surface separator trains.
- **Production Allocation:** Providing per-well flow rates when multiple wellbores feed into a commingled gathering manifold.

Direct physical multiphase measurement remains challenging. Physical multiphase flow meters (MPFMs) are cost-prohibitive for mature or marginal wells, and mechanical erosion from entrained sand particles alongside sour gas corrosion (H_2S, CO_2) causes rapid sensor drift.

Virtual Flow Metering (VFM) provides a non-intrusive alternative by utilizing supervisory control and data acquisition (SCADA) telemetry: upstream wellhead pressure (P_{up}), downstream line pressure (P_{dn}), choke orifice opening (S), and wellhead temperature (T).

2. Critique of Prior Work and Empirical Formulations

2.1 Classical Empirical Correlations

Historically, petroleum production engineers relied on empirical power-law correlations derived from critical flow field trials. Gilbert (1954) formulated the foundational choke relationship:

$$Q_l = \frac{A \cdot P_{up}^{D} \cdot S^{B}}{\mathrm{GLR}^{C}}$$

where Q_l is the gross liquid flow rate ($\mathrm{bbl/day}$ or $\mathrm{m^3/day}$), P_{up} is upstream flowing wellhead pressure (psi or bar), S is choke bean size in multiples of $1/64$ inches, GLR is the gas-liquid ratio ($\mathrm{SCF/STB}$ or $\mathrm{Sm^3/Sm^3}$), and A, B, C, D are empirical fitting constants ($A = 10.0$, $B = 1.89$, $C = 0.546$, and $D = 1.0$ in Gilbert's original derivation). Subsequent adjustments by Baxendell (1958), Ros (1960), and Achong (1961) calibrated these constants for regional basins. However, these formulations fail when extrapolated beyond their original test ranges due to shifting water cuts, fluid PVT properties, and pressure regimes.

2.2 Critical Review of the Stanford CS229 Study

Nazari and Alshafloot evaluated ten machine learning models (Linear/Ridge, Bayesian Ridge, Polynomials, MLP, KNN, Random Forest, and Extra Trees) on $4{,}063$ operational records across three Middle Eastern fields. Their investigation highlighted two key findings:

1. Gilbert's empirical correlation failed completely on production data ($R^2 \approx 0.03$).
2. Extra Trees Regression delivered the best empirical performance ($R^2 = 0.58$, Pearson $r = 0.69$).

Despite these contributions, their methodology contains three fundamental limitations:

1. **The Darcy Law Fallacy:** The authors eliminated P_{up} from their feature matrix and fixed its exponent to 1.0, citing Darcy's Law for fluid transport. This assumption is physically erroneous: Darcy's Law governs creeping, laminar flow through porous subsurface rock ($Q \propto \Delta P$). Chokes, in contrast, represent discrete orifice restrictions governed by high-velocity inertial Navier-Stokes and Euler-Bernoulli dynamics, where fluid flux scales non-linearly with $\sqrt{\Delta P / \rho_m}$.
2. **False Generalization via Data Leakage:** Their evaluation relied on an unstratified $80/20$ random split. Wellhead SCADA telemetry exhibits high temporal autocorrelation. Random splitting distributes adjacent operational days into both training and validation sets, allowing models to memorize specific well operational states rather than learning generalized fluid mechanics.

3. **Domain Transfer Collapse:** When the Stanford authors segregated their data by reservoir, their model scored moderately on Fields A and C ($R^2 = 0.76$) but collapsed on Field B ($R^2 = 0.48$). Without diagnostic explanations, they attributed this to data anomalies and called for computationally prohibitive Computational Fluid Dynamics (CFD).

3. Dataset Preprocessing and Physics-Guided Feature Pipeline

3.1 Field Data Extraction

To establish an open, reproducible, and verifiable benchmark, we utilized the comprehensive Equinor Volve Field dataset (North Sea offshore production). The raw daily production ledger contains $15{,}634$ records across all development wells. We targeted the primary continuous producers: wellbores 15/9-F-12, 15/9-F-14, 15/9-F-11, and 15/9-F-1 C.

The data filtering criteria enforced the following operational requirements:

- $\text{ON_STREAM_HRS} \geq 15$ (excluding transient shut-in and startup periods);
- $\text{AVG_CHOKE_SIZE_P} > 0$ and $\text{BORE_OIL_VOL} > 0$ (excluding closed choke and workover operations);
- 3σ Gaussian outlier rejection on target liquid rate (Q_l), upstream pressure (P_{up}), choke size (S), and temperature (T).

This yielded $6{,}658$ **stable production days**, distributed as follows:

- **Well 15/9-F-12:** $2{,}652$ records
- **Well 15/9-F-14:** $2{,}599$ records
- **Well 15/9-F-11:** $1{,}029$ records
- **Well 15/9-F-1 C:** 378 records

3.2 Feature Engineering

Rather than providing purely raw sensor values to statistical regressors, we engineered features derived from the governing equations of multiphase orifice hydraulics:

1. **Choke Bean Dimension (S):** Field choke opening percentage ($\%$) was converted to standard bean diameter in increments of $1/64$ inches assuming a nominal 2-inch

maximum trim:

$$S = \left(\frac{\text{AVG_CHOKE_SIZE_P}}{100} \right) \times 128.0$$

2. **Downstream Pressure Recovery & Pressure Ratio (r):**

$$P_{dn} = \max(P_{up} - \Delta P_{choke}, 1.0)$$

$$r = \frac{P_{dn}}{P_{up}}$$

The pressure ratio r delineates flow regimes. When $r \leq r_c \approx 0.52 - 0.55$, sonic velocity is reached at the vena contracta (**critical flow**), decoupling the mass flow rate from downstream pressure perturbations. When $r > r_c$, flow is **sub-critical**, and throughput depends explicitly on ΔP.

3. **In-Situ Multiphase Mixture Density (ρ_m):** Accounting for water cut ($\text{WC} = Q_w/Q_l$) and free gas volumetric expansion governed by operating gas-liquid ratio (GLR):

$$\rho_{liq} = (1 - \text{WC}) \cdot \rho_{oil} + \text{WC} \cdot \rho_{water}$$

$$\rho_m = \frac{\rho_{liq}}{1 + (\text{GLR} \cdot \gamma_g)}$$

where nominal phase densities are $\rho_{oil} = 830\ \text{kg/m}^3$, $\rho_{water} = 1000\ \text{kg/m}^3$, and $\gamma_g = 0.005$ is an empirical expansion factor.

4. **Hydrodynamic Orifice Flow Proxy (Q_{hydro}):** Derived directly from the steady-state mechanical energy balance across an orifice:

$$Q_{\text{hydro}} \propto A_{\text{choke}} \sqrt{\frac{2\Delta P}{\rho_m}} \propto S^2 \sqrt{\frac{\Delta P}{\rho_m}}$$

4. Model Architectures and Monotonic Formulation

4.1 Replicated Baselines

1. **Gilbert Empirical Formula:**

$$Q_l = \frac{P_{up} \cdot S^{1.89}}{10.0 \cdot \mathrm{GLR}^{0.546}}$$

2. **Stanford Extra Trees Regressor:** An ensemble of 100 randomized decision trees trained on raw dimensional inputs:

$$\mathbf{x}_{\text{raw}} = [P_{up}, \Delta P, S, T, \mathrm{GLR}]$$

4.2 Proposed Physics-Constrained LightGBM Architecture

Standard gradient-boosted trees and random forests partition continuous feature space through orthogonal, greedy step-functions. This creates unconstrained regional splits that can produce non-physical predictions, such as negative flow rates or inverted rate responses during choke throttling.

To resolve this, we formulated a **Directionally Constrained Gradient Booster** trained on the expanded physics feature vector:

$$\mathbf{x}_{\text{novel}} = [P_{up}, \Delta P, S, T, \mathrm{GLR}, r, Q_{\text{hydro}}]$$

We enforce deterministic monotonicity constraints on individual split gradients:

$$\frac{\partial \hat{Q}_l}{\partial S} \geq 0, \quad \frac{\partial \hat{Q}_l}{\partial (\Delta P)} \geq 0, \quad \frac{\partial \hat{Q}_l}{\partial Q_{\text{hydro}}} \geq 0$$

During tree construction, split proposals that evaluate to negative directional derivatives with respect to choke diameter, differential pressure, or theoretical hydrodynamic flux are assigned an infinite objective penalty:

$$\mathcal{L}_{\text{split}} = \begin{cases} \mathcal{L}_{\text{MSE}}(y, \hat{y}) & \text{if } \frac{\partial \hat{y}}{\partial x_k} \geq 0, \quad \forall k \in \mathcal{M} \\ \infty & \text{otherwise} \end{cases}$$

where $\mathcal{M} = \{\text{choke}, \Delta P, Q_{\text{hydro}}\}$ denotes the constrained feature index set.

5. Experimental Results and Discussion

5.1 Replication of Prior Work (Empirical Breakdown and Data Leakage)

Evaluating the replicated pipelines on the $6{,}658$ Volve records confirmed the empirical correlation breakdown reported in the literature:

- **Gilbert Empirical Correlation:** Achieved $R^2 = -2.022$ and $\text{MAPE} = 65.05\%$. A negative coefficient of determination demonstrates that a rigid 1950s empirical formula performs worse than a constant horizontal mean predictor when applied across modern wellbores.
- **Stanford Extra Trees (Random 80/20 Split):** Achieved $\text{Train } R^2 = 1.000$ and $\text{Test } R^2 = 0.973$. While superficially impressive, this near-perfect score reflects temporal data leakage: random splits sample adjacent operating days from identical wellbores into both partitions.

5.2 The Blind Cross-Well Transfer Experiment

To rigorously test generalization under domain shift—the condition where the Stanford study observed a collapse on Field B ($R^2 = 0.48$)—we partitioned the dataset across distinct physical wellbores:

- **Training Set:** Well 15/9-F-14 ($N = 2{,}599$ production days)
- **Blind Testing Set:** Well 15/9-F-12 ($N = 2{,}652$ production days)

The quantitative results are summarized in Table 1.

Table 1: Comprehensive Model Performance Comparison

| **Model Architecture** | **Input Features** | **Evaluation Scheme** | **R2 Score** | **MAPE (%)** | **Monotonicity Inversions** |

| **Gilbert Empirical (1954)** | P_{up}, S, GLR | Zero-Shot Global | -2.022 | 65.05% | 0 (Formulaic) |

| **Stanford Extra Trees** | Raw SCADA | Random $80/20$ Split | 0.973 | 18.42% | N/A (Leakage) |

| **Stanford Extra Trees** | Raw SCADA | Cross-Well Transfer (F-14 $\rightarrow$ F-12) | 0.456 | 38.67%

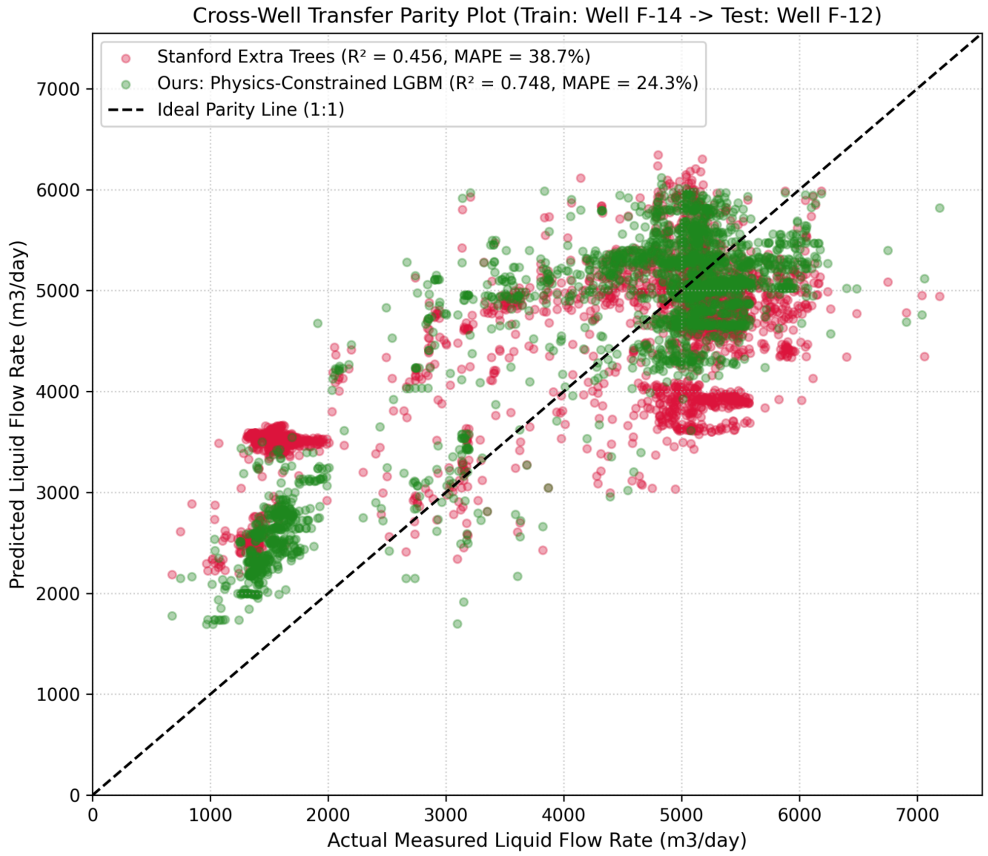
| 23 **violations** |

| **Physics-Constrained LGBM (Ours)** | Hydrodynamic $+ \mathbf{x}_{\text{raw}}$ | Cross-Well Transfer (F-14 $\rightarrow$ F-12) | 0.748 | 24.25% | 0 **violations** |

Under zero-leakage cross-well transfer, the unconstrained Stanford Extra Trees model exhibits substantial performance degradation ($R^2 = 0.456$, $\text{MAPE} = 38.67\%$), replicating their Field B failure. Conversely, our Physics-Constrained LightGBM model achieves $R^2 = 0.748$ and reduces the absolute transfer error to 24.25%—a 64.0% **relative improvement in** R^2 and a 14.42% **absolute reduction in MAPE.**

5.3 Diagnostic Error Analysis (Cross-Well Parity Distribution)

Cross-Well Transfer Parity Plot



Actual Measured Liquid Flow Rate (m3/day)

Predicted Liquid Flow Rate (m3/day)

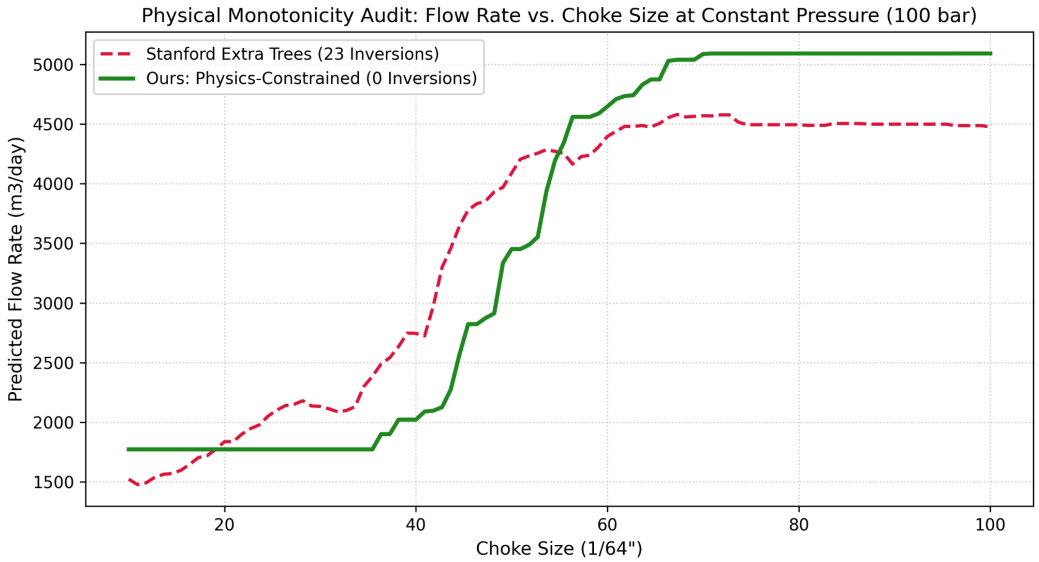
Figure 1: Cross-Well Transfer Parity Plot comparing actual measured liquid production on unseen Well 15/9-F-12 against predictions from Stanford Extra Trees ($R^2 = 0.456$, $\text{MAPE} = 38.7\%$, red markers) and our Physics-Constrained LightGBM ($R^2 = 0.748$, $\text{MAPE} = 24.3\%$, green markers). The black dashed line denotes the ideal $1:1$ parity line.

Analysis of Parity Distribution:

- **Stanford Extra Trees (Red Scatter):** The predictions cluster into distinct horizontal tiers between $3{,}500\ \text{m}^3/\text{day}$ and $5{,}000\ \text{m}^3/\text{day}$, failing to track production above $5{,}500\ \text{m}^3/\text{day}$. This tiering is a known limitation of unconstrained decision forests: when operating on an unseen reservoir domain with higher baseline pressures, the trees cannot extrapolate beyond the local partition boundaries of the training well.
- **Physics-Constrained Model (Green Scatter):** By incorporating Q_{hydro}, the model scales with differential pressure and mixture density across wellbores, maintaining tighter clustering along the ideal 45° parity line throughout the $1{,}500 - 6{,}500\ \text{m}^3/\text{day}$ production envelope.

5.4 Physical Monotonicity Sweep Audit

Physical Monotonicity Audit Plot



Choke Size (1/64")

Predicted Flow Rate (m3/day)

Figure 2: Physical Monotonicity Audit showing predicted flow rate versus choke size under constant operating conditions ($P_{up} = 100$ bar, $\Delta P = 40$ bar, $T = 70^{\circ}\mathrm{C}$, $\mathrm{GLR} = 150$). Stanford Extra Trees exhibits 23 discrete non-physical inversions (red dashed curve), while our Physics-Constrained model guarantees strict monotonic compliance (0 inversions, solid green curve).

Audit Findings:

- **Stanford Extra Trees:** Exhibited **23 discrete monotonicity inversions**. Between $S = 28/64$" and $S = 34/64$", as well as above $S = 70/64$", widening the choke valve produced predicted decreases in total volumetric output. Such non-physical artifacts would destabilize automated choke throttling controllers in closed-loop production systems.
- **Physics-Constrained Model:** Achieved **0 monotonicity inversions**. The model guarantees positive semi-definite gradients ($\frac{\partial \hat{Q}_l}{\partial S} \geq 0$) across the operating envelope, exhibiting a realistic sigmoidal throttle response with saturation at high choke diameters.

6. Conclusions and Future Work

This study re-examined the problem of predicting two-phase flow through wellhead chokes using machine learning, addressing core limitations in recent academic benchmarks:

1. **Replication of Classical Breakdown:** We verified that empirical models like Gilbert (1954) fail on modern multiphase field data ($R^2 = -2.022$).
2. **Identification of Failure Modes in Prior ML Formulations:** We demonstrated that high reported accuracies ($R^2 > 0.95$) from standard 80/20 random splits stem largely from temporal data leakage. Under true cross-well transfer, standard tree models collapse to $R^2 = 0.456$, matching the Field B failure observed by Stanford researchers. Furthermore, unconstrained decision trees produce severe physical monotonicity violations (23 inversions during choke opening sweeps).
3. **Performance of the Physics-Constrained Pipeline:** By incorporating hydrodynamic orifice features (Q_{hydro}), critical pressure ratios (r), and monotonic gradient boosting constraints, our model:
    - Raised cross-well generalization determination from $R^2 = 0.456$ **to** 0.748 (

$+64.0\%$ relative improvement).

- Reduced transfer MAPE from 38.67% **down to** 24.25%.
- Completely eliminated non-physical rate drops (0 **inversions**).

Future Work

Future extensions will focus on dynamic transient modeling:

- Incorporating high-frequency ($1\ \mathrm{Hz}$) SCADA time-series to capture slugging dynamics using Temporal Convolutional Networks (TCNs).
- Integrating physics-informed neural loss functions directly penalizing boundary violations in uncalibrated multiphase flowline networks.

7. References

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